Modelling with Parametric Equations

Exam code: 9MA0

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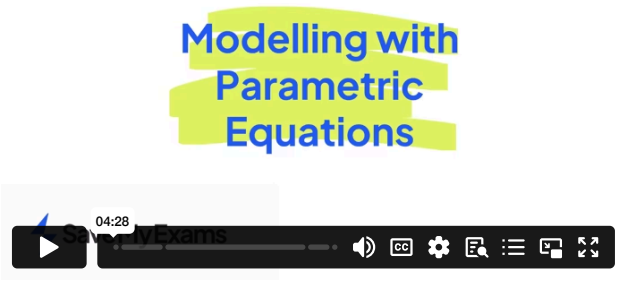
Written by: Paul

Updated on 16 February 2026



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Modelling with parametric equations

What can be modelled with parametric equations?

- A common use for parametric equations is motion in two dimensions
  - **x = f(t)** – horizontal position, dependent on time
  - **y = g(t)** – vertical position, dependent on time
- This is explored in vectors and projectiles

A stone is launched from a catapult at ground level. Relative to the origin, where the catapult is situated, the position of the stone, at time t seconds, is defined parametrically by

$x = t^2 + t$
$y = 8t - 2t^2$ where x and y are measured in metres

a) find the position of the stone after 1 second
b) at which times is the stone at a height of 6 metres
c) assuming the ground is horizontal, how far away from the catapult does the stone first hit the ground after launch?
d) i) find the time at which the stone is at its maximum height
ii) find the coordinates of the stone when it is at its maximum height
e) what can you deduce about the path of the stone from your answers above?

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Got it

$y = 8 - 2 = 6$

$\therefore$ AFTER 1 SECOND STONE IS AT $(2, 6)$

b) HEIGHT OF 6m $\rightarrow y = 6$

$8t - 2t^2 = 6$

$t^2 - 4t + 3 = 0$

$(t-1)(t-3) = 0$

$t = 1, \quad t = 3$

STONE IS AT 6m AFTER 1 SECOND AND 3 SECONDS

c) GRAND LEVEL, HEIGHT OF 0m $\rightarrow y = 0$

$8t - 2t^2 = 0$

$2t(4 - t) = 0$

$\cancel{t = 0} \quad t = 4$

AT START STONE ON CATAPULT!

THIS IS THE TIME, QUESTION WANTS DISTANCE

$x = 4^2 + 4$

$\therefore x = 20$ m

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d) i) MAXIMUM HEIGHT – HORIZONTAL TANGENT

$\frac{dy}{dt} = 8 - 4t = 0 \quad$ WHEN $t = 2$

$\frac{d^2y}{dt^2} = -4 < 0 \quad \therefore$ MAXIMUM

NOT ESSENTIAL UNLESS ASKED FOR

STONE IS AT MAXIMUM HEIGHT AFTER 2 SECONDS

ii) AT $t = 2, \quad x = 2^2 + 2 = 6$

$y = 16 - 8 = 8$

COORDINATES OF STONE AT MAXIMUM HEIGHT IS $(6, 8)$

e) THE STONE DOES NOT FOLLOW A SYMMETRICAL PATH – IF IT DID WE'D EXPECT MAX HEIGHT TO BE AT $x = 10$m

HALF OF 20m WHEN IT HITS GROUND

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Worked Example

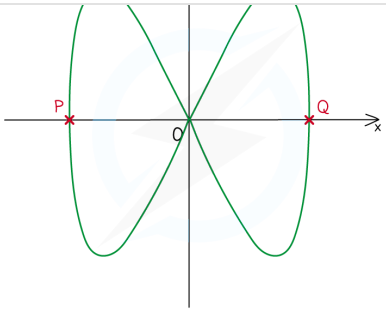
A model car, C, travels along a model track along the path of the curve shown in the diagram below and defined by the parametric equations

$$x = 2\cos t$$

$$y = 3\sin 2t$$

where $0 \le t \le 2\pi$

x and y are measured in metres, t is measured in minutes.



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(a) Find the initial position of the car, C

(b) Find the times at which $(0 \le t \le 2\pi)$ the car is at the origin

(c) Find the coordinates of the point labelled X on the diagram

(d) At which other times $(0 \le t \le 2\pi)$ is C the same distance from the origin as it is when it is at point X?

(e) Find the distance PQ on the diagram

(f) If the restriction $0 \le t \le 2\pi$ was removed from the model what could you say about the motion of the model car?

a) INITIAL POSITION, $t = 0$

$x = 2\cos 0 = 2$

$y = 3\sin 0 = 0$

$\therefore$ INITIAL POSITION OF C IS $(2, 0)$

b) ORIGIN POSITION $(0, 0)$

$2\cos t = 0$

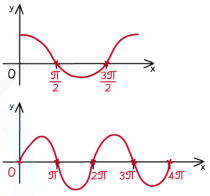
$t = \frac{\pi}{2}, \frac{3\pi}{2}$

$3\sin 2t = 0$

$2t = 0, \pi, 2\pi, 3\pi, 4\pi$

$t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$

MUST HAVE BOTH x & y EQUAL TO 0



$\therefore$ C IS AT ORIGIN WHEN $t = \frac{\pi}{2}$ AND $t = \frac{3\pi}{2}$ MINUTES

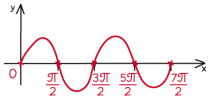
c) AT x, HORIZONTAL TANGENT, $\frac{dy}{dt} = 0$

$\frac{dy}{dt} = 6\cos 2t$

$\cos 2t = 0$

$2t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$

$t = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$



x AND y ARE BOTH POSITIVE AT x SO THIS MUST BE WHEN $t = \frac{\pi}{4}$

AT $t = \frac{\pi}{4}$, $x = 2\cos\left(\frac{\pi}{4}\right) = \frac{2}{\sqrt{2}} = \sqrt{2}$

$y = 3\sin\left(\frac{\pi}{2}\right) = 3$

$\therefore$ $x(\sqrt{2}, 3)$

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